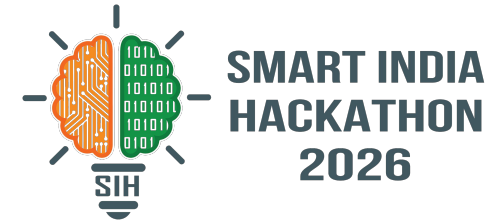


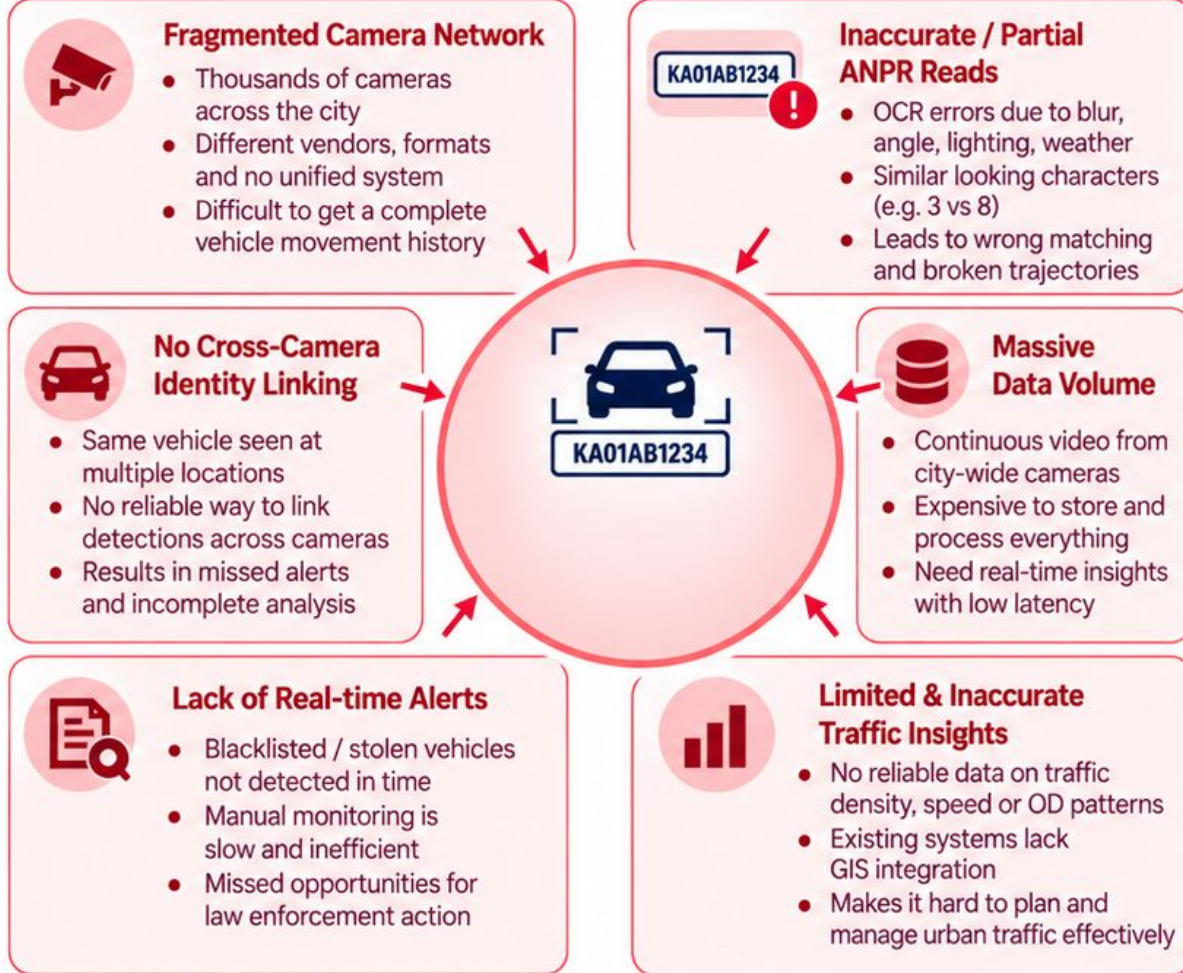
SMART INDIA HACKATHON 2026



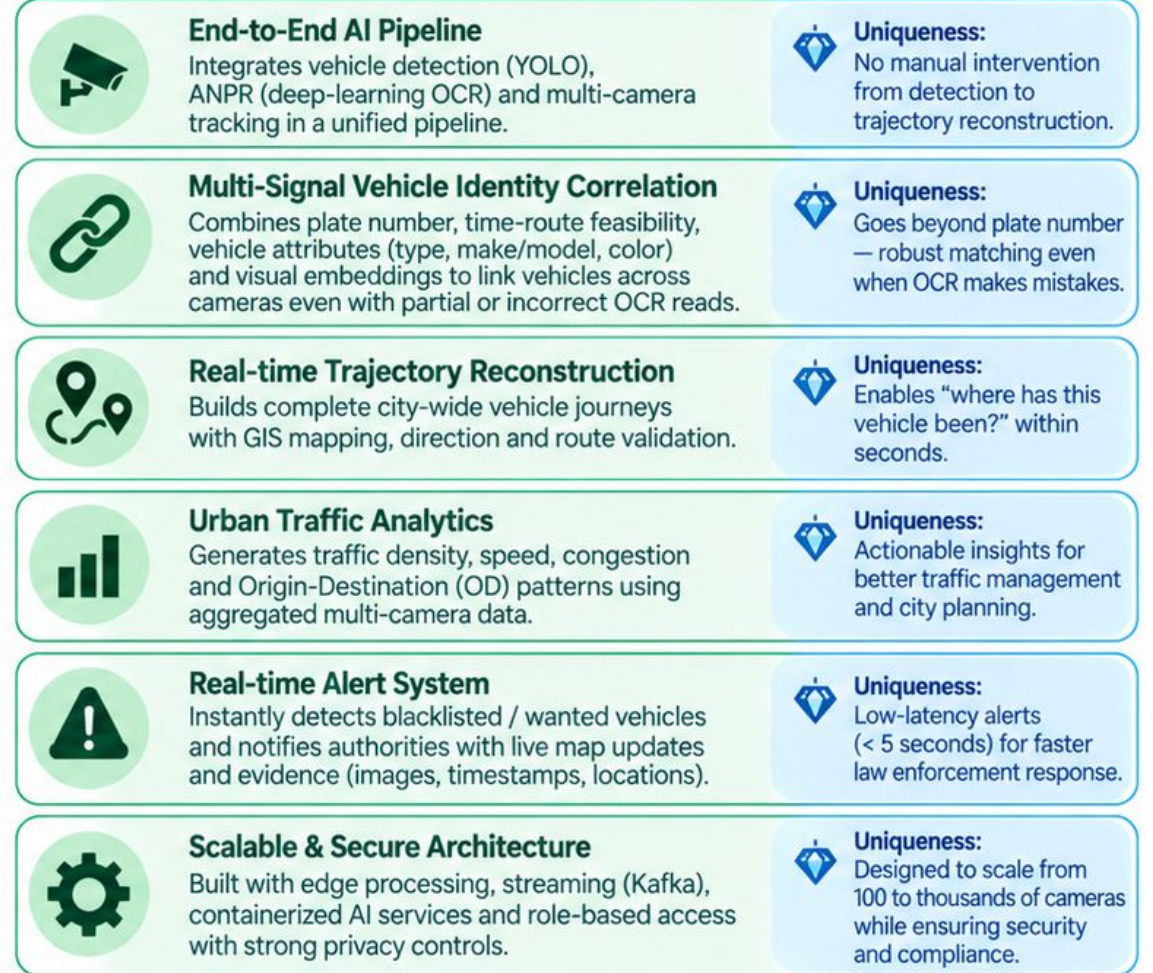
- **Problem Statement ID** – SIH26127
- **Problem Statement Title** -
City-Wide AI Engine for Multi-Camera
ANPR Trajectory Tracking and Urban
Traffic Analytics
- **Theme** - Smart Automation
- **PS Category**- Software
- **Team ID**-
- **Team Name** - Trajectory Titans

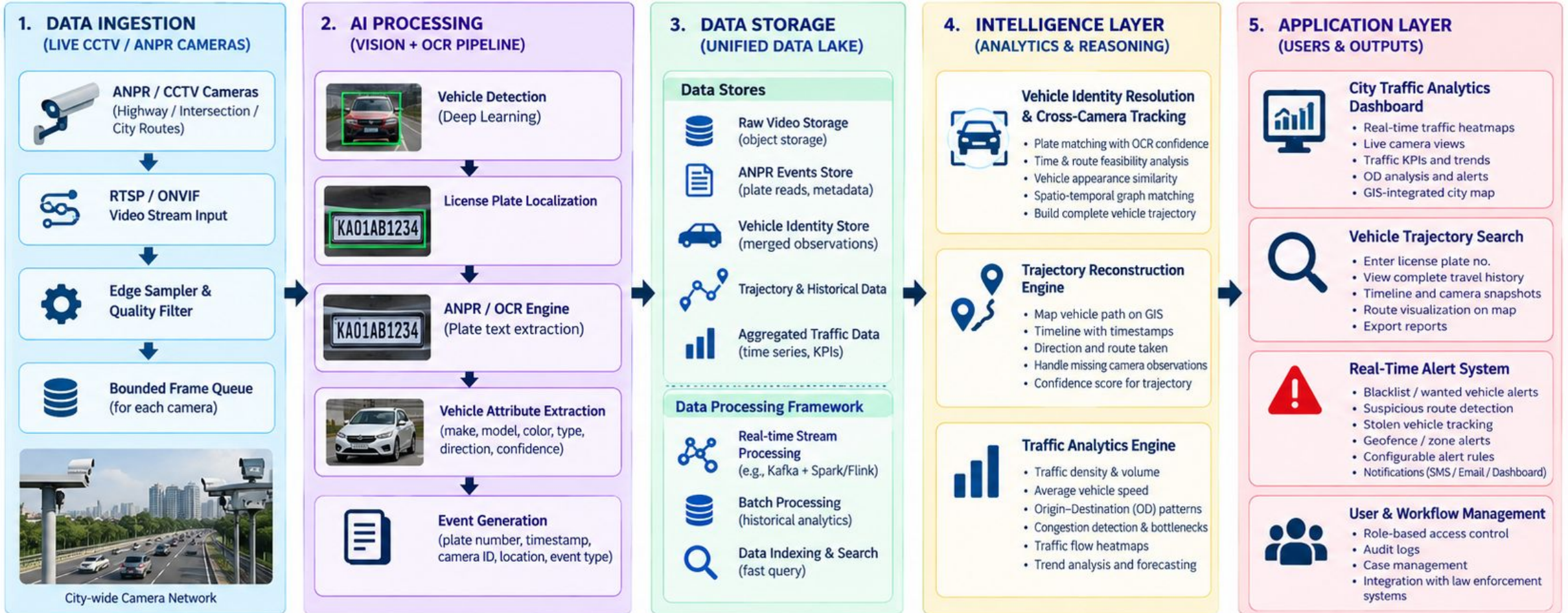


Understanding of Problems



Solution Strategy





TECH STACK

AI / ML

YOLOv8

OpenCV

EasyOCR

DeepSORT

Backend / Processing

Python

FastAPI

Kafka

Spark / Flink

PostgreSQL

Docker

Frontend / Visualization

React

MapLibre GL

WebSocket

Grafana



Understanding of Problems

Key issues in real-world deployment

01

OCR Misreads Corrupt the Trail

A misread plate from a different vehicle can silently join the trajectory.

02

No Indian Multi-Camera Dataset

Indian plate data and cross camera tracking data don't exist together publicly.

03

Cross-Camera Clock Drift

Hundreds of cameras drifting out of sync corrupts trajectory ordering.

04

City-Scale Video Volume

Processing and storing every frame from every camera is prohibitively expensive.

05

Camera Heterogeneity

Different vendors, resolutions, protocols and metadata formats across the network.

06

Surveillance Misuse Risk

Plate and movement data is sensitive; unrestricted access invites abuse.

07

Climate Challenges

Adverse weather (rain, fog, dust, glare) and low-light conditions degrade plate visibility and tracking accuracy.



Solution Strategy

Our approach for a practical and scalable solution

01

Grammar + Feasibility Filtering

Per state HSRP decode constraints, confusion weighted matching, OSRM travel-time validation.

02

Decoupled Data Strategy

Indian sets and synthetic HSRP train OCR; CityFlowV2 trains cross-camera tracking independently.

03

NTP/PTP Synchronisation

Centralised timestamp validation with continuous drift monitoring on every camera node.

04

Edge-First Processing

Sample and detect at the camera; store events and metadata, archive only evidence-grade video.

05

Camera Abstraction Layer

RTSP/ONVIF layer normalising streams, metadata and capabilities behind one interface.

06

Privacy by Design

7-day retention, case-ID bound audit-logged queries, RBAC, and encryption, DPDP Act 2023 aligned.

07

Multi-frame + IR / Low-light Support

Multi-frame capture, IR/low-light camera support, and vehicle appearance embedding to maintain robust tracking in adverse weather and poor lighting.



Social Impact

Safer, more efficient and more liveable cities

01

Faster emergency response

Cross-camera tracking speeds up police action.

02

Fewer road accidents

Automated violation detection, quicker enforcement.

03

Greater public trust

Visible, data-backed enforcement builds confidence.

04

Aids missing-person cases

A vehicle linked to a case becomes traceable citywide.



Economic Impact

Cost-efficient, scalable and sustainable urban management

01

Lower enforcement cost

Less manual footage review, smaller monitoring teams.

02

Recovers lost revenue

Toll and fine evasion across routes becomes traceable.

03

Cuts congestion costs

Traffic analytics optimise signals, saving time and fuel.

04

No new hardware cost

Built on existing CCTV — zero capex for new cameras.

01



Detection

[YOLOv11 \(Ultralytics\)](#)

Detects vehicles in real-time with high accuracy and speed, used as the primary object detector.

02



OCR (default)

[PaddleOCR](#)

Recognizes license plate text from detected vehicle images, robust to different fonts, angles and lighting conditions.

03



Within-camera tracking

[ByteTrack \(arXiv 2110.06864\)](#)

Tracks vehicles across frames within the same camera, maintaining consistent identities even under occlusion.

04



Feasibility routing

[OSRM](#)

Validates camera-to-camera movement using real road network distances and travel time, ensuring physically feasible trajectories.

05



Cross-camera Re-ID

[TransReID](#)

Uses vehicle appearance embeddings to match the same vehicle across different cameras, even when the license plate is unreadable.

